Statistics 651
Advanced MCMC

Winter 2026

BYU
Stats

Objectives

1. Improve efficiency of MCMC via transformation and auxiliary variables
2. Construct slice samplers
3. Implement slice steps within a Gibbs sampler
4. Survey other advanced algorithms

Resources

- BDA3:
  - Ch 12.3–12.4
- BIDA: Ch 6.3.4

Transformation

Efficiency of Gibbs sampling

Gibbs sampling is sensitive to the topology of the target distribution.

It works best when elements of $\boldsymbol{\theta}$ are (nearly) uncorrelated.

Example: Negative Binomial model

Poisson models are inadequate for overdispersed count data.

A common alternative is Negative Binomial, for which $\mathbf{Var}(Y) \geq \mathbb{E}(Y)$.

The mass function is

$$p(y \mid r, \phi) = \frac{\Gamma(y + r)}{\Gamma(r)\, y!} \phi^r (1 - \phi)^y$$

for $y = 0, 1, 2, \ldots$, $r > 0$ and $0 < \phi \leq 1$.

Transformation

Example (continued): Negative Binomial model

Consider count data (sorted)

```
y <- c(42, 44, 46, 48, 48, 48, 51, 54, 59, 63, 73, 74, 74, 77, 80)
```

with mean 58.7 and variance 182.9,

and the model

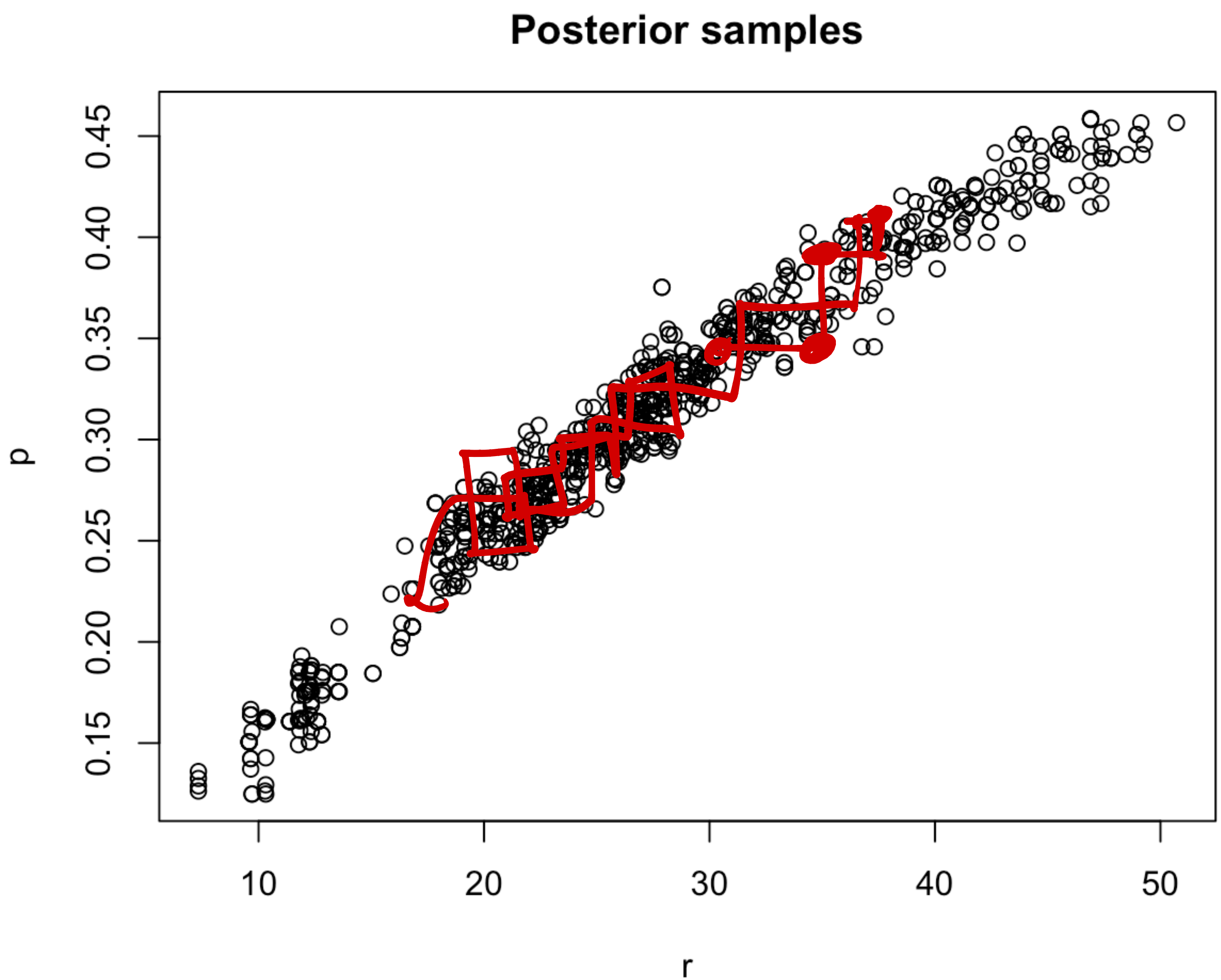
$$
\begin{aligned}
y_i \mid r, p &\overset{\text{iid}}{\sim} \text{NegBinom}(r, p),\ i = 1, \ldots, 15, \\
r \sim \text{Gamma}(2, \texttt{scale} = 10), &\quad p \sim \text{Beta}(1, 1)\,.
\end{aligned}
$$

Sample from the joint posterior of $\boldsymbol{\theta} = (r, p)$ using a Gibbs sampler.

▶ Code

Transformation

Example (continued): Negative Binomial model

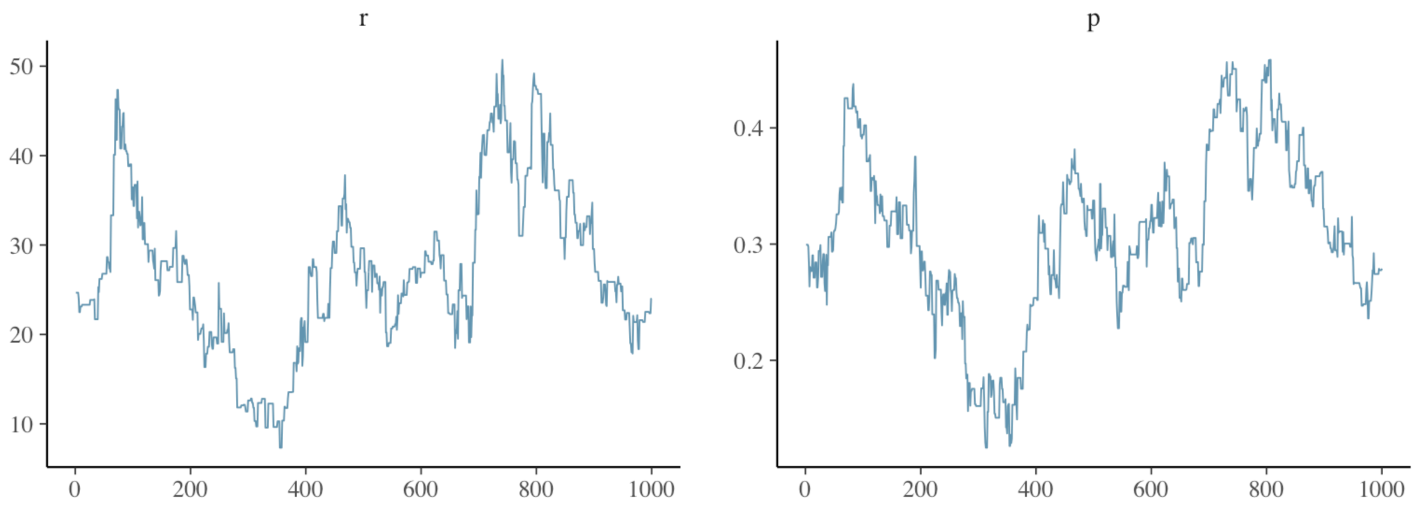


Posterior samples
p
0.15
0.20
0.25
0.30
0.35
0.40
0.45
r
10
20
30
40
50

Transformation

Example (continued): Negative Binomial model

```r
bayesplot::mcmc_trace(mcmc_out$sims)
```




```r
posterior::ess_bulk(mcmc_out$sims[,"r"])
```

```
[1] 7.80757
```

```r
posterior::ess_bulk(mcmc_out$sims[,"p"])
```

```
[1] 7.71139
```

Transformation

Example (continued): Negative Binomial model

Now fit the same model, but sample instead

$$\mu = \mathbb{E}(Y) = \frac{r(1-p)}{p}$$

$$\mu = \frac{r\left(1-\frac{1}{cv}\right)}{\frac{1}{cv}} = r(cv-1) = \mu \Rightarrow r = \frac{\mu}{cv-1}$$

and the coefficient of variation,

$$\text{cv} = \frac{\text{Var}(Y)}{\mu}, \text{ where } \text{Var}(Y) = \frac{r(1-p)}{p^2}.$$

$$cv = \frac{\frac{r(1-p)}{p^2}}{\frac{r(1-p)}{p}} = \frac{1}{p} \Rightarrow p = \frac{1}{cv}$$

Don't forget to sample the *transformed* target posterior!

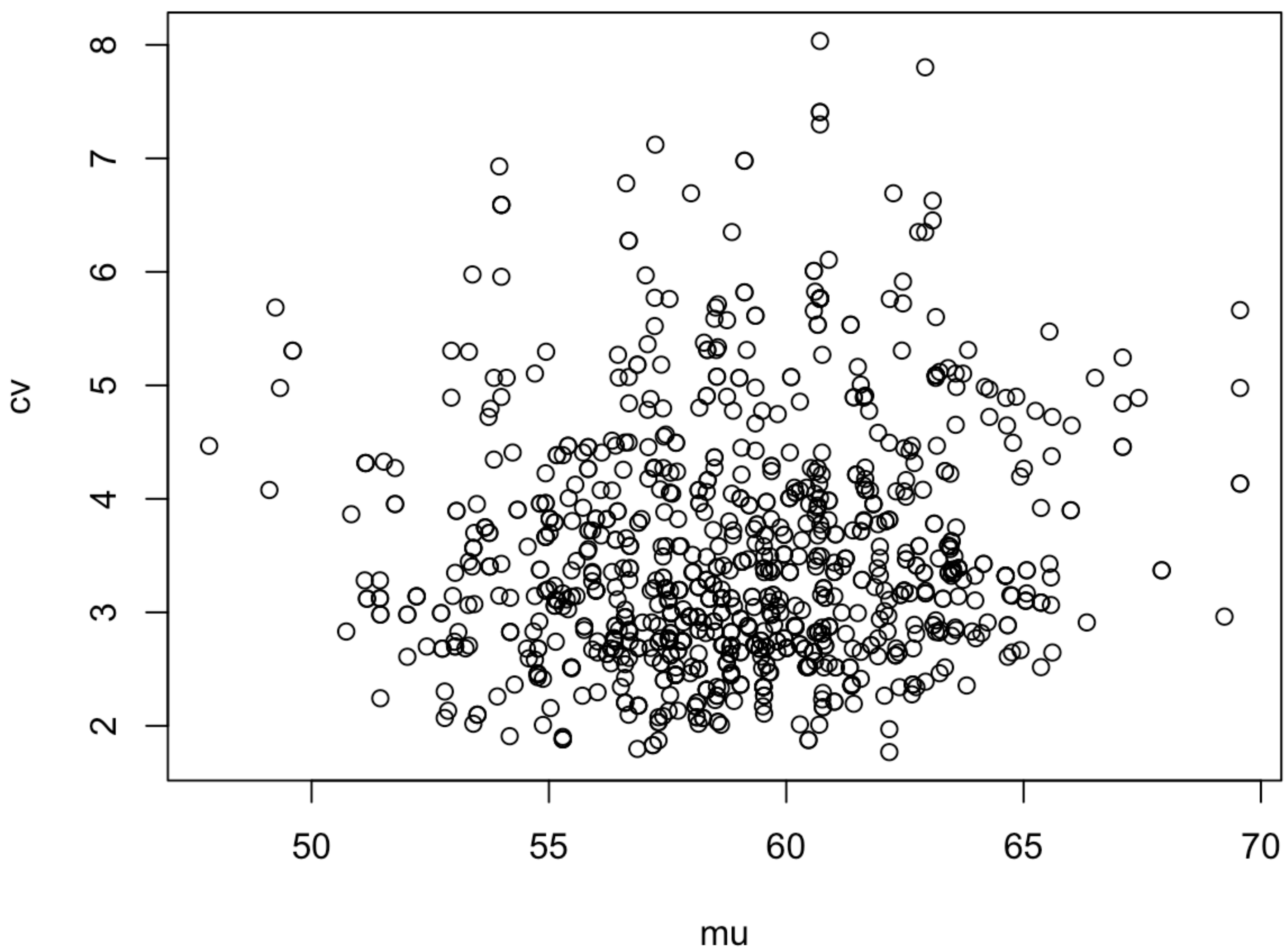
▶ Code

$$\begin{pmatrix} \frac{\partial r}{\partial \mu} & \frac{\partial r}{\partial cv} \\ \frac{\partial p}{\partial \mu} & \frac{\partial p}{\partial cv} \end{pmatrix} = \begin{pmatrix} \frac{1}{cv-1} & \text{forget about it... not needed} \\ 0 & -\frac{1}{cv^2} \end{pmatrix}$$

Transformation

Example (continued): Negative Binomial model

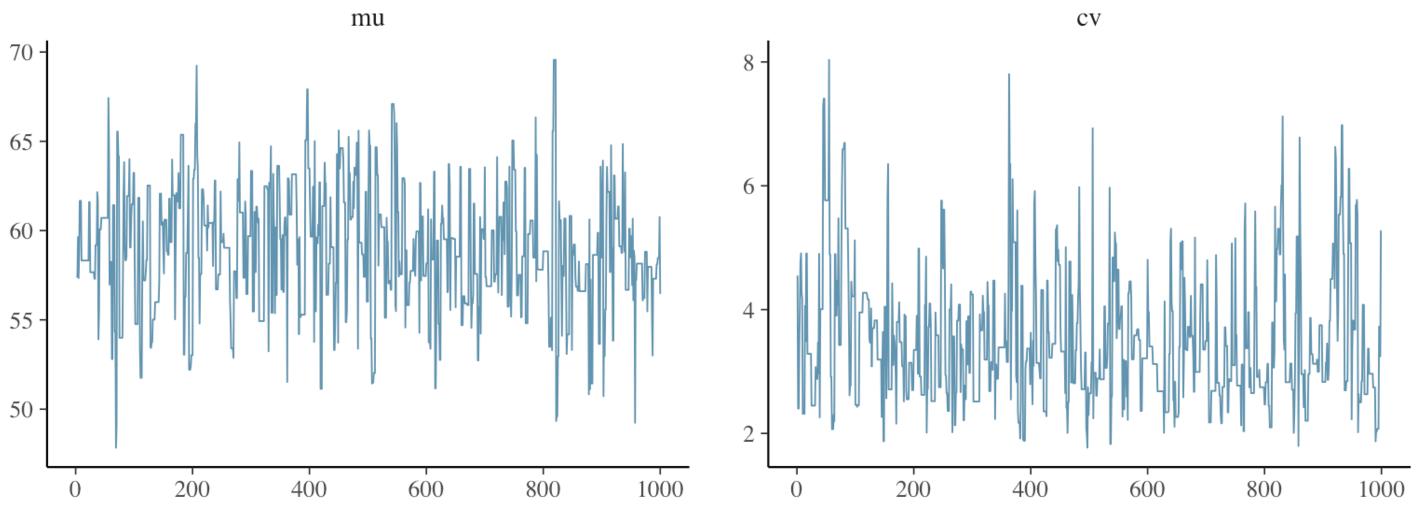
Posterior samples



Transformation

Example (continued): Negative Binomial model

```r
bayesplot::mcmc_trace(mcmc_out2$sims)
```




```r
posterior::ess_bulk(mcmc_out2$sims[,"mu"])
```

```
[1] 320.1535
```

```r
posterior::ess_bulk(mcmc_out2$sims[,"cv"])
```

```
[1] 191.1429
```

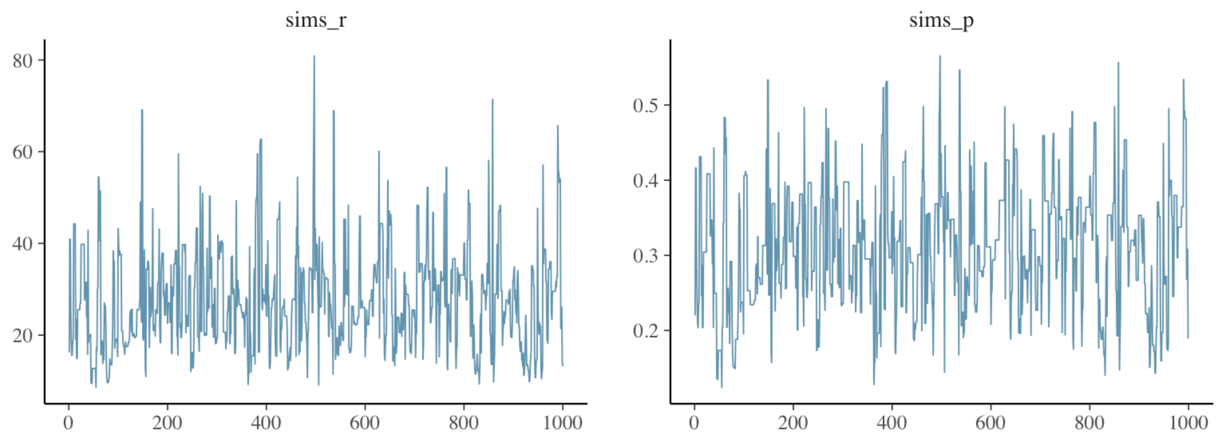
Transformation

Example (continued): Negative Binomial model

If we really care about the parameters r and p, we can back-transform

```
sims_r <- mcmc_out2$sims[,"mu"] / (mcmc_out2$sims[,"cv"] - 1.0)
sims_p <- 1.0 / mcmc_out2$sims[,"cv"]

bayesplot::mcmc_trace(cbind(sims_r, sims_p))
```




```
posterior::ess_bulk(sims_r)
```

```
[1] 202.2383
```

```
posterior::ess_bulk(sims_p)
```

```
[1] 191.1429
```

Transformation

Example (continued): Negative Binomial model

We specified priors for r and p,

but if we care about μ and $\mathbf{cv}$ (which are more interpretable), why not place priors directly on those in the first place?

Restricted support

Common transformation for positive-valued parameters

If $\theta > 0$, consider transforming to sample $\psi = \log(\theta) \in \mathbb{R}$.

Inverse transformation

$$\theta = e^{\psi}$$

Jacobian

$$e^{\psi}$$

So if the original target density on θ were $\pi(\theta \mid \text{data})$, your target density for ψ would be

$$\pi(\theta \mid \text{data}) \times e^{\psi}$$

plug in $h^{-1}(\psi) = e^{\psi}$ for θ

Restricted support

Common transformation for bounded parameters

If $0 < \theta < 1$, consider transforming to sample $\phi = \log\left(\frac{\theta}{1-\theta}\right) \in \mathbb{R}$.

Inverse transformation

$$\theta = \frac{1}{1 + e^{-\phi}}$$

Jacobian

$$\frac{e^{-\phi}}{(1 + e^{-\phi})^2}$$

Back-transforming

Sample from the (transformed) posterior of

$$\psi = h(\theta)$$

to obtain $(\psi^{(1)}, \ldots, \psi^{(S)})$[1].

Transforming back,

$$\theta^{(s)} \leftarrow h^{-1}(\psi^{(s)})$$

yields $\theta^{(s)} \sim$ original target.

Auxiliary variables (augmentation)

Auxiliary variables

Non-conjugate models can sometimes be made *conditionally conjugate* (or just easier to sample) by introducing **latent (auxiliary) variables** such that the original target Π is recovered via marginalization

$$\begin{aligned}\Pi(\boldsymbol{\theta}) &= \int \Pi_0(\boldsymbol{\theta}, \mathrm{d}\boldsymbol{\psi}) \\ &= \int p(\boldsymbol{\theta} \mid \boldsymbol{\psi})\, p(\boldsymbol{\psi})\, \mathrm{d}\boldsymbol{\psi}\,.\end{aligned}$$

It is sometimes easier (or more efficient) to build a Gibbs sampler for the **augmented** parameter vector $(\boldsymbol{\theta}, \boldsymbol{\psi})$ that targets Π_0.

Auxiliary variables

Example: t distribution

The Student-t distribution with ν degrees of freedom, location μ, and scale σ

can be written as a **scale mixture** of normals. That is,

$f_Y(y) = \int N(y|v) IG(v)\, dv$

$Y \sim t_\nu(\mu, \sigma)$

$t_\nu(\mu, \sigma)$

is the marginal distribution of Y from

$$
\begin{aligned}
Y \mid V &\sim \text{Normal}(\mu, V)\,, \\
V &\sim \text{Inv-Gamma}\left(\frac{\nu}{2}, \frac{\nu\sigma^2}{2}\right)\,.
\end{aligned}
$$

When modeling data $y_i \stackrel{\text{iid}}{\sim} t_v(\mu, \sigma)$, the augmentation introduces auxiliary (and unobserved) $v_i \stackrel{\text{iid}}{\sim} \text{Inv-Gamma}$ for each observation $i = 1, \ldots, n$.

Ch 12.1 of BDA3 gives details under prior $(\mu,\ \log(\sigma)) \sim \text{Uniform}$.

Auxiliary variables

Example: Half-Cauchy distribution

$$IG(x \mid a, b) = \frac{b^a}{\Gamma(a)} x^{-a-1} e^{-\frac{b}{x}}$$

The prior $\pi(\sigma) = \text{Cauchy}^+(\sigma;\ s_0)$ is equivalent to

$$\sigma^2 \mid \eta \sim \text{Inv-Gamma}\left(\frac{1}{2}, \frac{1}{\eta}\right),$$
$$\eta \sim \text{Inv-Gamma}\left(\frac{1}{2}, \frac{1}{s_0^2}\right).$$

Can you show that we recover $\pi(\sigma)$ by averaging over η?

$$p(\sigma^2, \eta) = \frac{\left(\frac{1}{\eta}\right)^{\frac{1}{2}}}{\Gamma\left(\frac{1}{2}\right)} (\sigma^2)^{-\frac{1}{2}-1} \exp\left[-\frac{1}{\eta\sigma^2}\right] \frac{\left(\frac{1}{S_0^2}\right)^{\frac{1}{2}}}{\Gamma\left(\frac{1}{2}\right)} (\eta)^{-\frac{1}{2}-1} \exp\left[-\frac{1}{S_0^2 \eta}\right]$$

$$= \frac{(\sigma^2)^{-1}}{\pi S_0 \sigma} \eta^{-1-1} \exp\left[-\frac{1}{\eta}\left(\frac{1}{S_0^2} + \frac{1}{\sigma^2}\right)\right]$$

$$p(\sigma^2) = \int \text{above}\, d\eta = \frac{(\sigma^2)^{-1}}{\pi S_0 \sigma} \frac{1}{\left(\frac{1}{S_0^2} + \frac{1}{\sigma^2}\right)} = \frac{1}{\pi S_0 \sigma} \frac{1}{\left(\frac{\sigma^2}{S_0^2} + 1\right)}$$

let $y = \sqrt{\sigma^2}$

$\sigma^2 = y^2 \qquad J = 2y$

$$p(\sigma) = \frac{1}{\pi s_0 \sigma}\left(1 + \frac{\sigma^2}{s_0^2}\right)^{-1} 2\sigma$$

$$= \frac{2}{\pi s_0}\left(1 + \frac{\sigma^2}{s_0^2}\right)^{-1} \quad \text{for } \sigma > 0$$

Auxiliary variables

Example: Half-Cauchy distribution

Complete conditional distributions under $y_i \overset{\text{iid}}{\sim} \mathrm{Normal}(\mu, \sigma^2)$ for $i = 1, \ldots, n$ using $\pi(\sigma) = \mathrm{Cauchy}^+(\sigma;\ s_0)$:

$$\begin{aligned}
\sigma^2 \mid \eta,\, \mu,\, \boldsymbol{y} &\sim \text{Inv-Gamma}\left(\frac{n+1}{2}, \frac{\sum_{i=1}^n (y_i - \mu)^2}{2} + \frac{1}{\eta}\right), \\
\eta \mid \sigma^2,\, \mu,\, \boldsymbol{y} &\sim \text{Inv-Gamma}\left(1, \frac{1}{\sigma^2} + \frac{1}{s_0^2}\right).
\end{aligned}$$

Auxiliary variables

Example: zero-inflated Poisson model

The model

$$\mathbb{P}(y_i = 0 \mid \pi, \lambda) = \pi + (1 - \pi)\exp(-\lambda)\,,$$

$$\mathbb{P}(y_i = k \mid \pi, \lambda) = (1 - \pi)\exp(-\lambda)\lambda^k/k!\,, \quad \text{for } k = 1, 2, \ldots$$

can be rewritten as a mixture pmf:

$$p(y_i \mid \pi, \lambda) = \pi\, 1\{y_i = 0\} + (1 - \pi)\,\mathrm{Pois}(y_i; \lambda)\,.$$

Now write a hierarchical model with latent z_i indicating whether y_i comes from the first mixture component.

$z_i = 1$ if y_i is from first component (0)

Hier model

$$z_i \overset{iid}{\sim} \mathrm{Bern}(\pi)$$

$$y_i \mid z_i \overset{iid}{\sim} \begin{cases} \mathbb{1}(y_i = 0) & \text{if } z_i = 1 \\ \mathrm{Pois}(\lambda) & \text{if } z_i = 0 \end{cases}$$

Auxiliary variables

Example: zero-inflated Poisson model

Hierarchical model:

$$z_i \stackrel{\text{iid}}{\sim} \text{Bern}(\pi) \text{ for } i = 1, \ldots, n,$$

$$(y_i \mid z_i = 1) \sim 1\{y_i = 0\} \quad \text{and} \quad (y_i \mid z_i = 0) \stackrel{\text{ind}}{\sim} \text{Pois}(\lambda)\,.$$

Can we recover the original pmf by integrating z_i out?

$$P(y_i, z_i) = f(y_i \mid z_i)\, P(z_i)$$

$$= \left[\mathbb{1}(y_i = 0)\, \mathbb{1}(z_i = 1) + \frac{e^{-\lambda} \lambda^{y_i}}{y!} \mathbb{1}(z_i = 0) \right] \pi^{z_i} (1 - \pi)^{1 - z_i}$$

$$P(y_i) = \sum_{z_i = 0}^{1} P(y_i, z_i) = \frac{e^{-\lambda} \lambda^{y_i}}{y_i} (1 - \pi) + \pi\, \mathbb{1}(y_i = 0)$$

Auxiliary variables

Example: zero-inflated Poisson model

Hierarchical model:

$$z_i \overset{\text{iid}}{\sim} \text{Bern}(\pi) \text{ for } i = 1, \dots, n,$$
$$(y_i \mid z_i = 1) \sim 1\{y_i = 0\} \quad \text{and} \quad (y_i \mid z_i = 0) \overset{\text{ind}}{\sim} \text{Pois}(\lambda)\,.$$

Now consider priors $\pi \sim \text{Beta}(\alpha, \beta)$ and $\lambda \sim \text{Gamma}(\kappa, \theta)$ with $\alpha = \beta = \theta = 1$ and $\kappa = 2$.

What are the complete conditional distributions for z_i, π, and λ?

$$f(\text{everything}) = \prod_{i=1}^{n} \left[\left[1(y_i=0)\, 1(z_i=1) + \frac{e^{-\lambda}\lambda^{y_i}}{y_i!} 1(z_i=0) \right] \pi^{z_i} (1-\pi)^{1-z_i} \right] \times$$
$$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \pi^{\alpha-1} (1-\pi)^{\beta-1} \frac{\theta^{\kappa}}{\Gamma(\kappa)} \lambda^{\kappa-1} e^{-\theta\lambda}$$

$$\propto \prod_{i: z_i=1} \left[\pi\, 1(y_i=0) \right] \prod_{i: z_i=0} \left[\frac{e^{-\lambda}\lambda^{y_i}}{y_i!} (1-\pi) \right] \pi^{\alpha-1} (1-\pi)^{\beta-1} \lambda^{\kappa-1} e^{-\theta\lambda}$$

FC for π: $\propto \pi^{\sum z_i}(1-\pi)^{n-\sum z_i}\,\pi^{\alpha-1}(1-\pi)^{\beta-1}$

$$\propto Be\left(\alpha+\sum z_i,\ \beta+n-\sum z_i\right)$$

FC for λ: $\propto \prod_{i:z_i=0}\left[e^{-\lambda}\lambda^{y_i}\right]\lambda^{k-1}e^{-\theta\lambda}$

$$= e^{-\lambda(n-\sum z_i)}\,\lambda^{\sum_{z_i=0} y_i}\,\lambda^{k-1}\,e^{-\theta\lambda}$$

$$\propto Ga\left(k+\sum_{i:z_i=0} y_i,\ \text{rate}=\theta+n-\sum z_i\right)$$

FC for each z_i:

$$\pi \mathbb{1}(y_i=0)\mathbb{1}(z_i=1) + \frac{e^{-\lambda}\lambda^{y_i}}{y_i!}(1-\pi)\mathbb{1}(z_i=0)$$

$$P(z_i=1 \mid \cdots) \propto \pi \mathbb{1}(y_i=0) = w_{1i}$$

$$P(z_i=0 \mid \cdots) \propto \frac{e^{-\lambda}\lambda^{y_i}}{y_i!}(1-\pi) = w_{2i}$$

$$z_i \mid \cdots \sim \text{Bern}\left(\frac{w_{1i}}{w_{1i}+w_{2i}}\right)$$

Slice sampling

Slice sampling

Fundamental Theorem of Simulation

Simulating a random variable θ from a distribution having density (or mass) function $f(\theta)$ is equivalent to

1. simulating from $f(\theta, v) = 1\{0 < v < f(\theta)\}$, and
2. discarding V.

Ch 2.3.1 of Robert and Casella (2004) (Ch 4.6 in 1999 edition available online through the BYU Library).

Here V is an auxiliary variable.

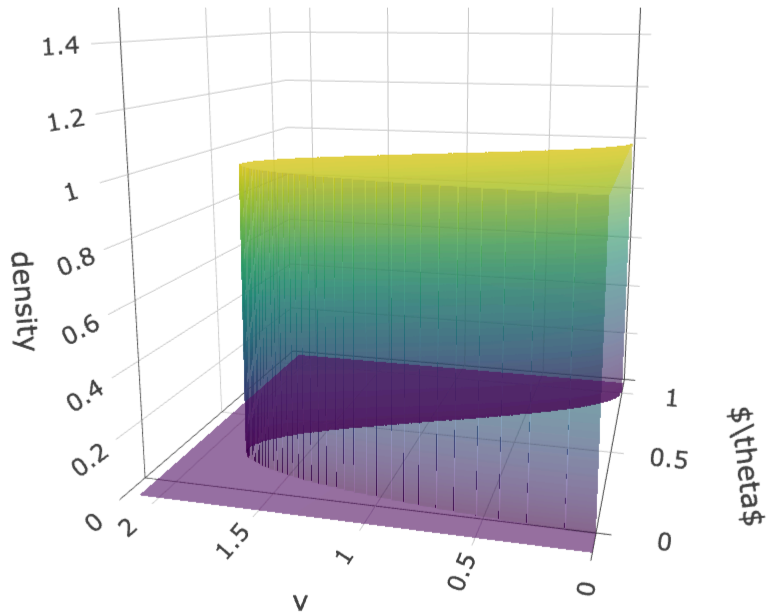
Why is this true?

$$f(\theta, v) = 1(0 < v < f(\theta))$$

$$\int_0^\infty 1(0 < v < f(\theta))\, dv = \int_0^{f(\theta)} 1\, dv = f(\theta)$$

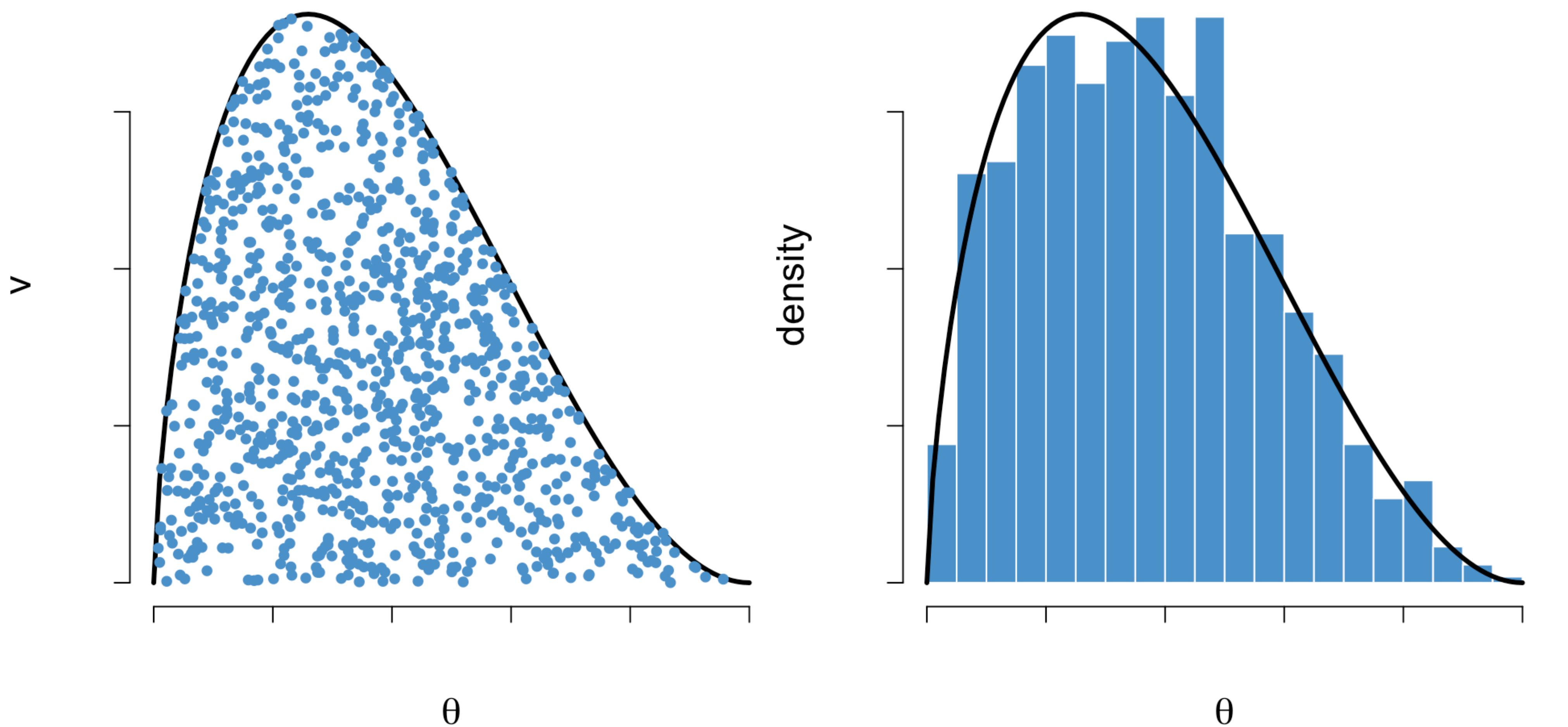
Slice sampling

Fundamental Theorem of Simulation: $f(\theta, v) = 1\{0 < v < f(\theta)\}$



Slice sampling

Fundamental Theorem of Simulation: $f(\theta, v) = 1\{0 < v < f(\theta)\}$



Slice sampling

Simple slice sampler

Suppose we have target density $\pi(\theta) = c\,g(\theta)$, but $c > 0$ is unknown.

We can still sample the augmented joint density $\pi(\theta, v) = c\,1\{0 < v < g(\theta)\}$ via Gibbs sampling.

What are the two complete conditional distributions?

Full conditional of v:

$$v \mid \theta \sim \text{Unif}(0, g(\theta))$$

Full condition of θ:

$$\theta \mid v \sim \text{Unif}(\{\theta : g(\theta) > v\})$$

Slice sampling

Simple slice sampler

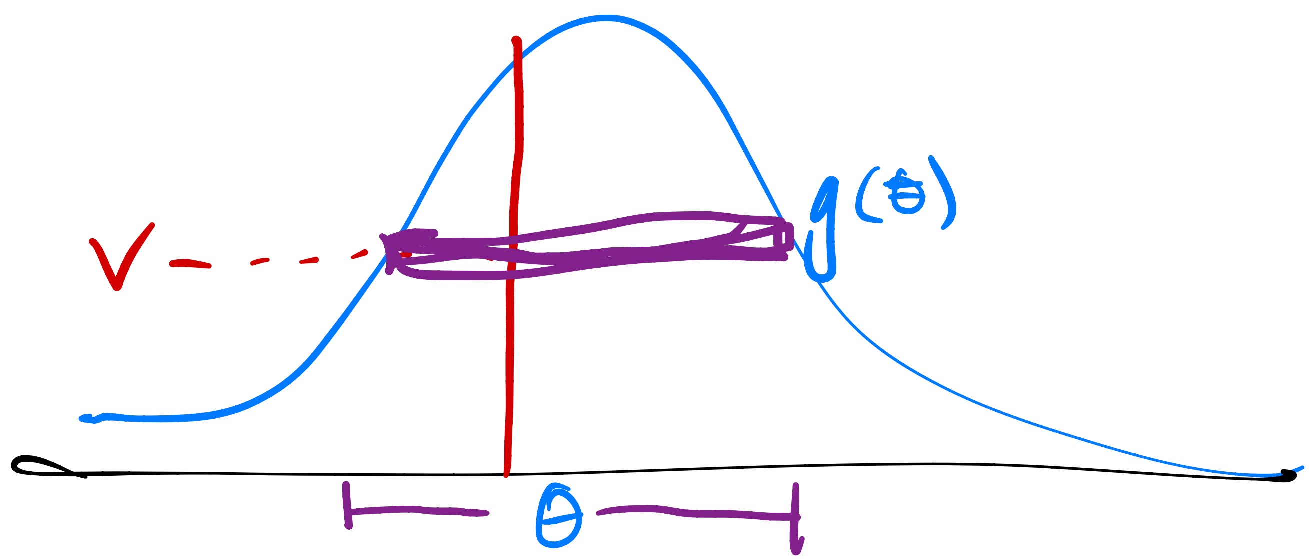
Sample the augmented joint density $\pi(\theta, v) = c\,1\{0 < v < g(\theta)\}$.

One sweep can also be thought of as a transition from θ' to θ:

1. $V \mid \theta' \sim \mathrm{Uniform}(0, g(\theta'))$
2. $\theta \mid V = v \sim \mathrm{Uniform}(A(v))$

where $A(v) := \{\theta : v < g(\theta)\}$ is called the **slice region**.

Transition kernel:



Simple slice sampler

Problem: how can we sample uniformly over the set $\{\theta : v < g(\theta)\}$?

Example

Develop a simple slice sampler for a standard normal target distribution.

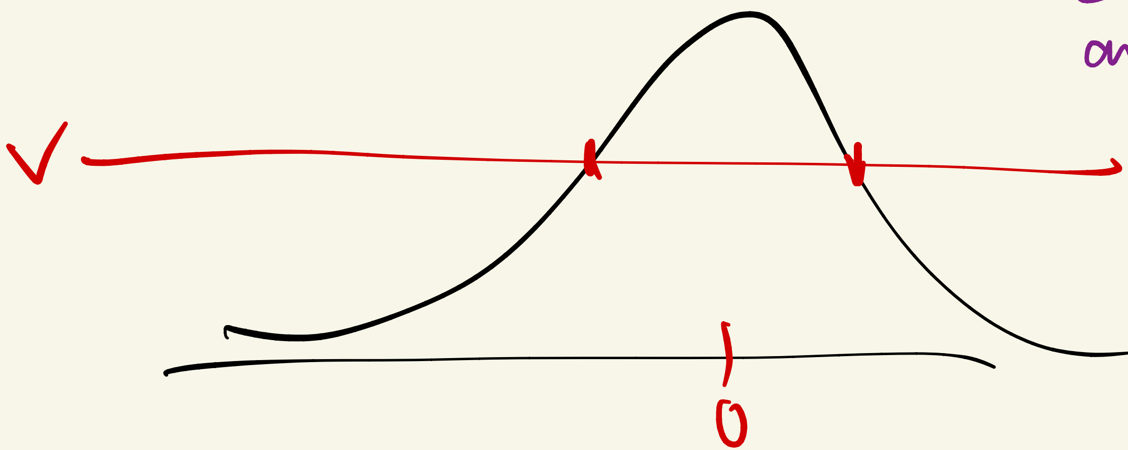
$$g(x) = e^{-\frac{x^2}{2}}$$

1. draw $V \mid X \sim \text{Unif}\left(0, e^{-x^2/2}\right)$
2. draw $X \mid V \sim \text{Unif}\big(\underbrace{\{x : e^{-x^2/2} > v\}}_{\text{slice region}}\big)$

$$e^{-x^2/2} > v \iff -x^2/2 > \log v$$

$$\iff x^2 < -2 \log v$$

$$-\sqrt{-2\log V} < x < +\sqrt{-2\log V}$$



since $e^{-x^2/2}$ never gets higher than $e^{-0^2/2} = 1$, $V<1$ and so

$$-2\log V > 0$$

always. Thus we will always have a non-complex solution in this case.

Simple slice sampler

Problem: how can we sample uniformly over the set $\{\theta : v < g(\theta)\}$?

What if we can't solve for this set?

Simple slice sampler

Problem: how can we sample uniformly over the set $\{\theta : v < g(\theta)\}$?

What if we can't solve for this set?

General solution

Neal (2003) proposes a hybrid algorithm that introduces two *nested* procedures that preserve the uniform distribution over the slice region.

1. Stepping out (Figure 3 in the paper)
2. Shrinkage (Figure 5 in the paper)

Both procedures are implemented in the R package qslice:

`qslice::slice_stepping_out()`

Illustration of shrinkage procedure

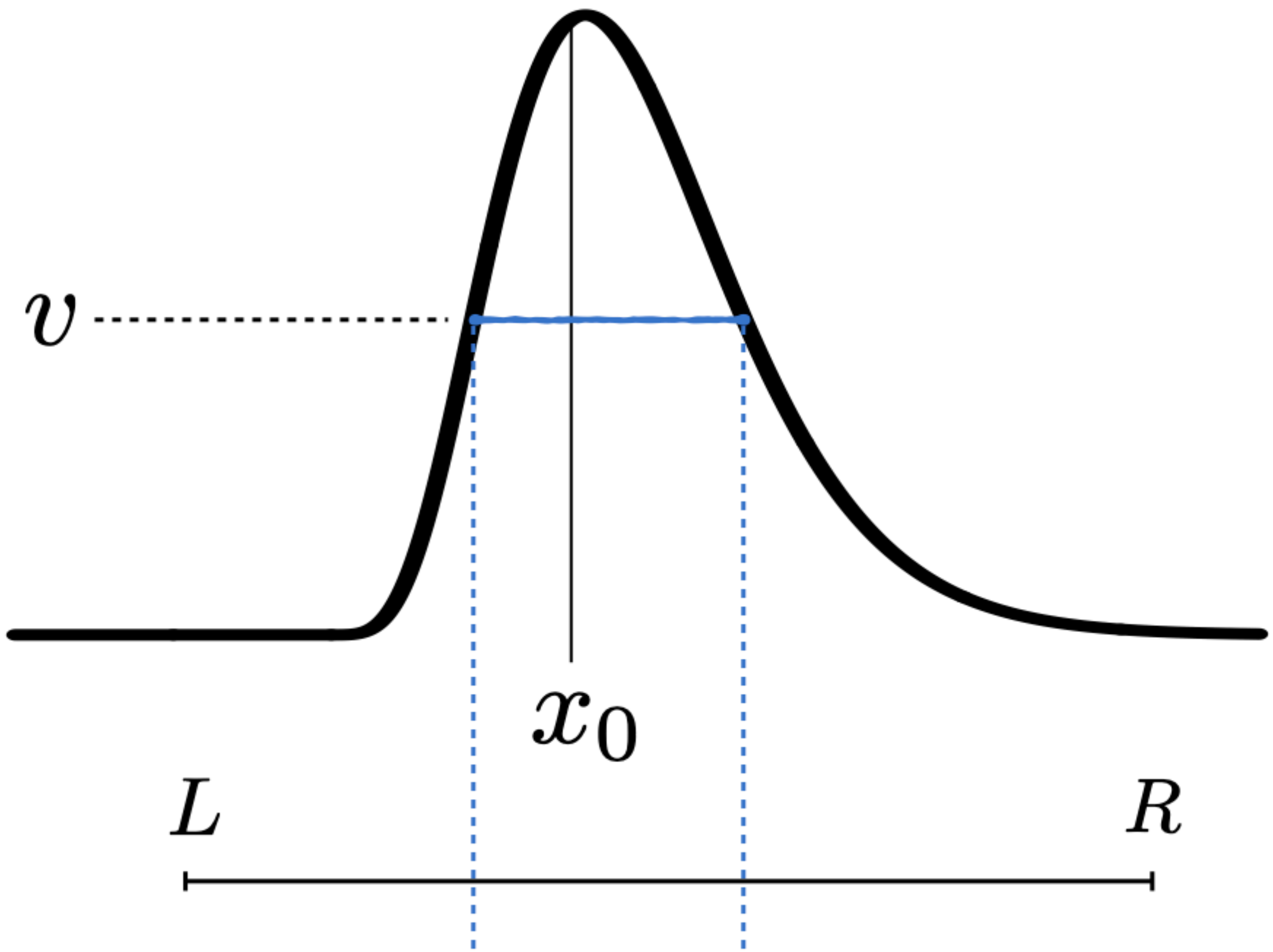


Illustration of shrinkage procedure

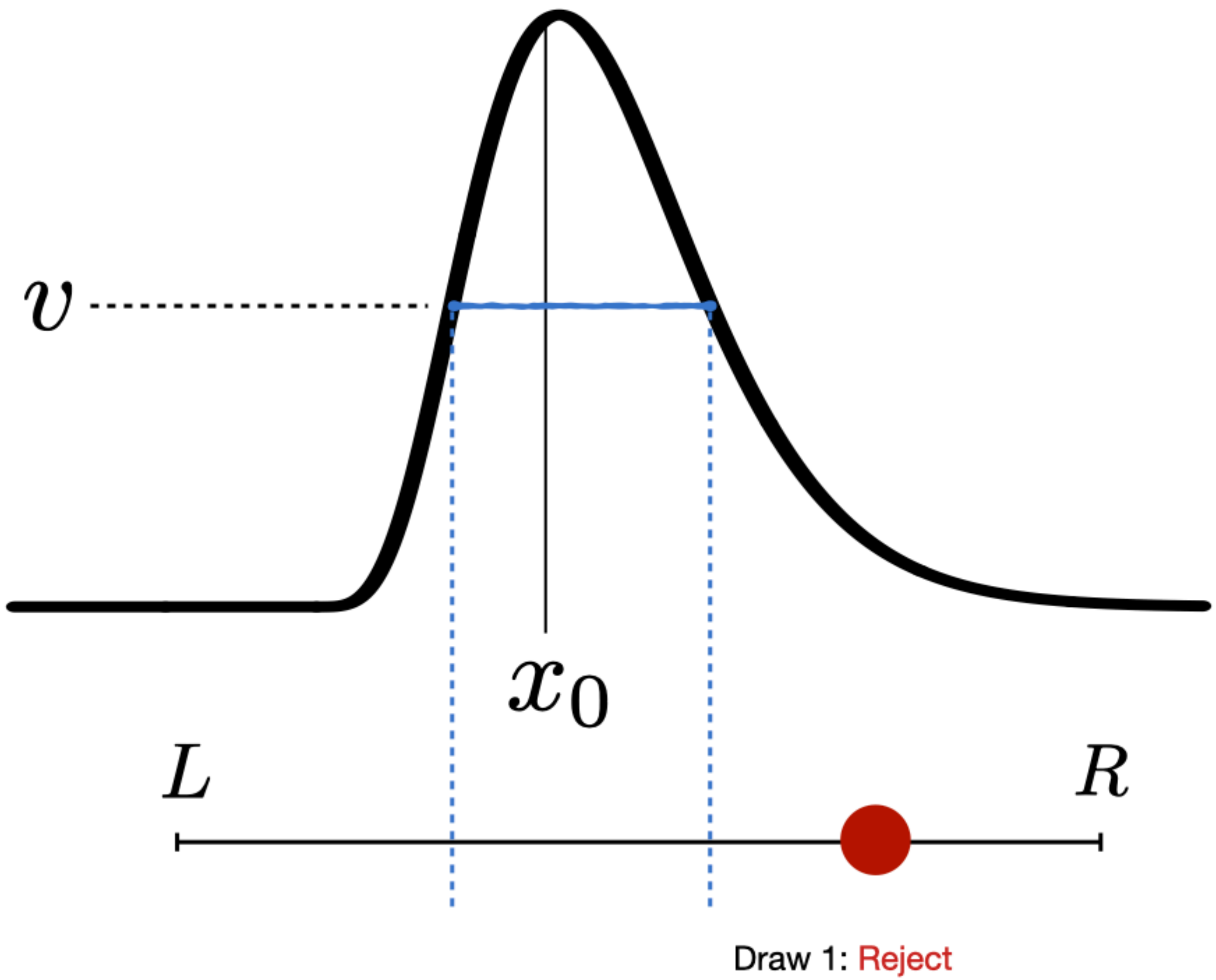


Illustration of shrinkage procedure

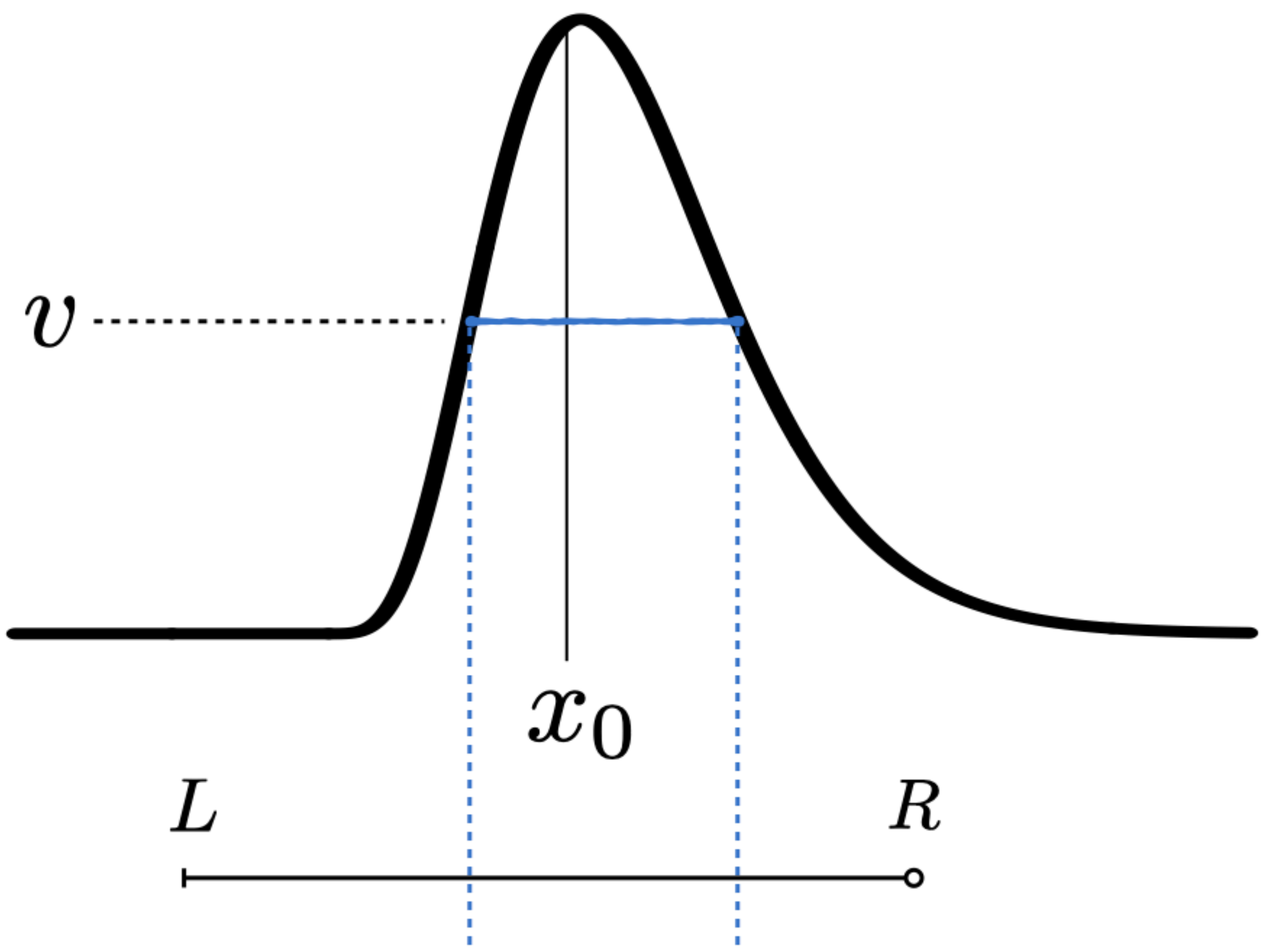
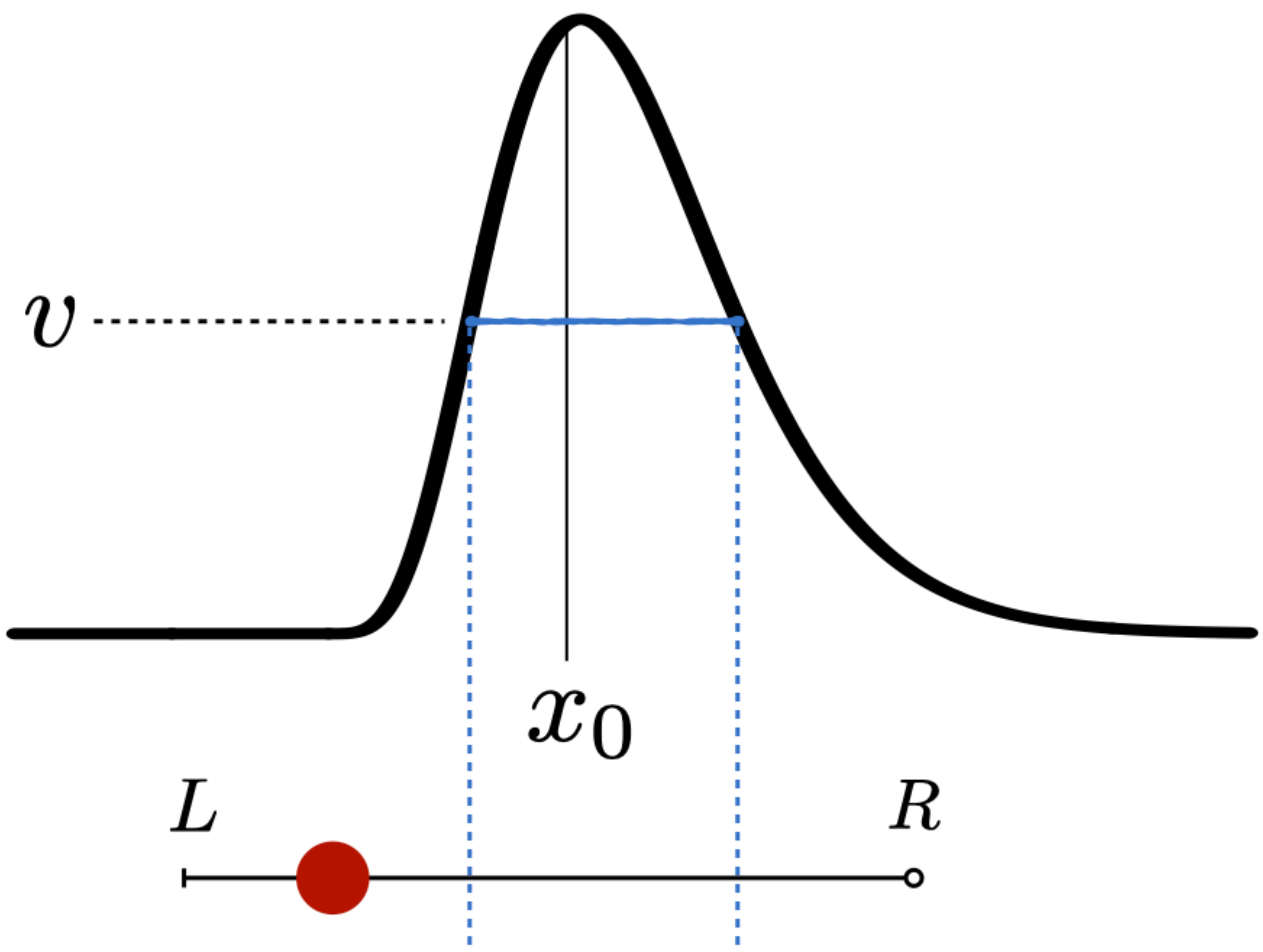


Illustration of shrinkage procedure



Draw 2: Reject

Illustration of shrinkage procedure

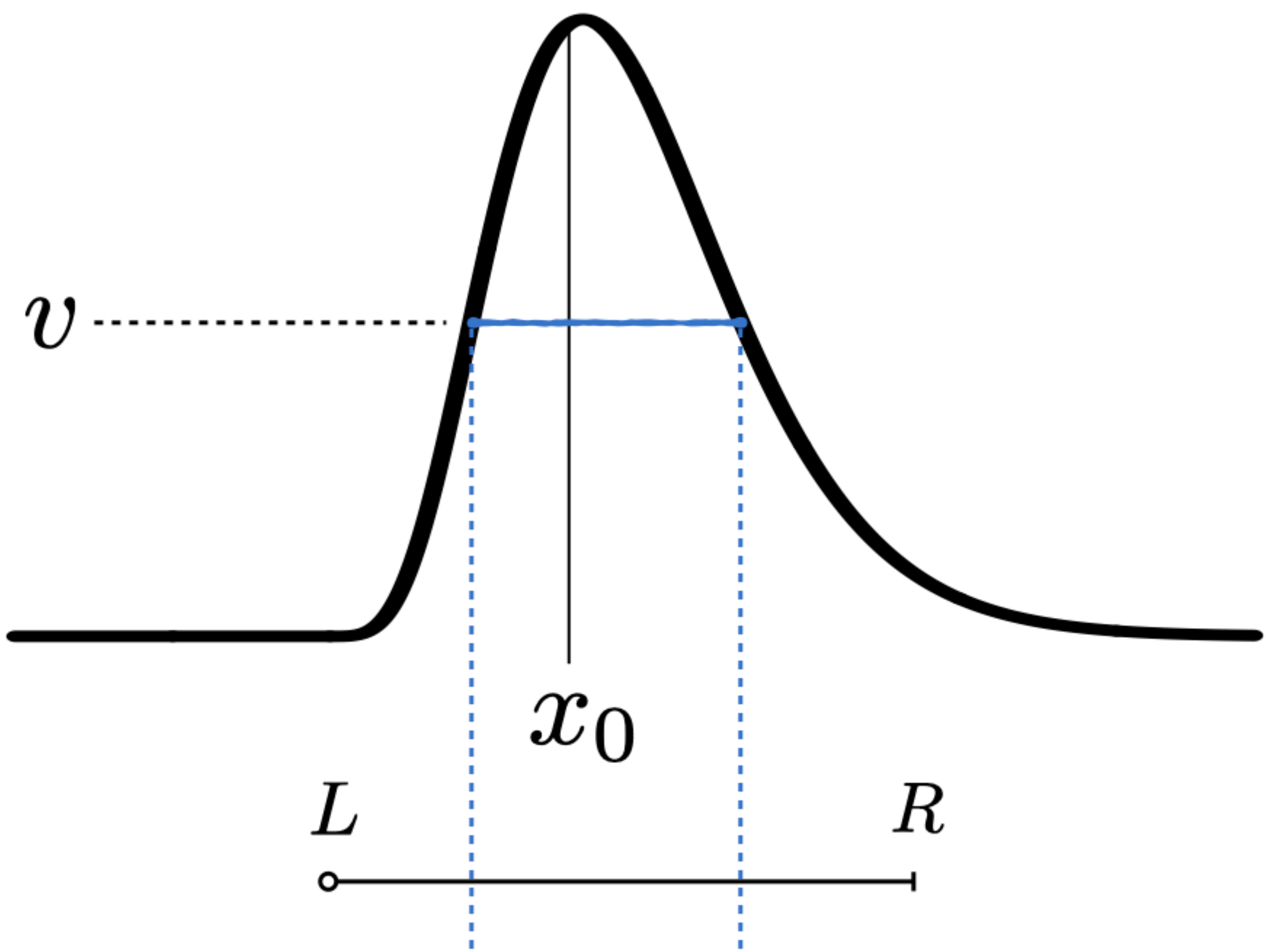
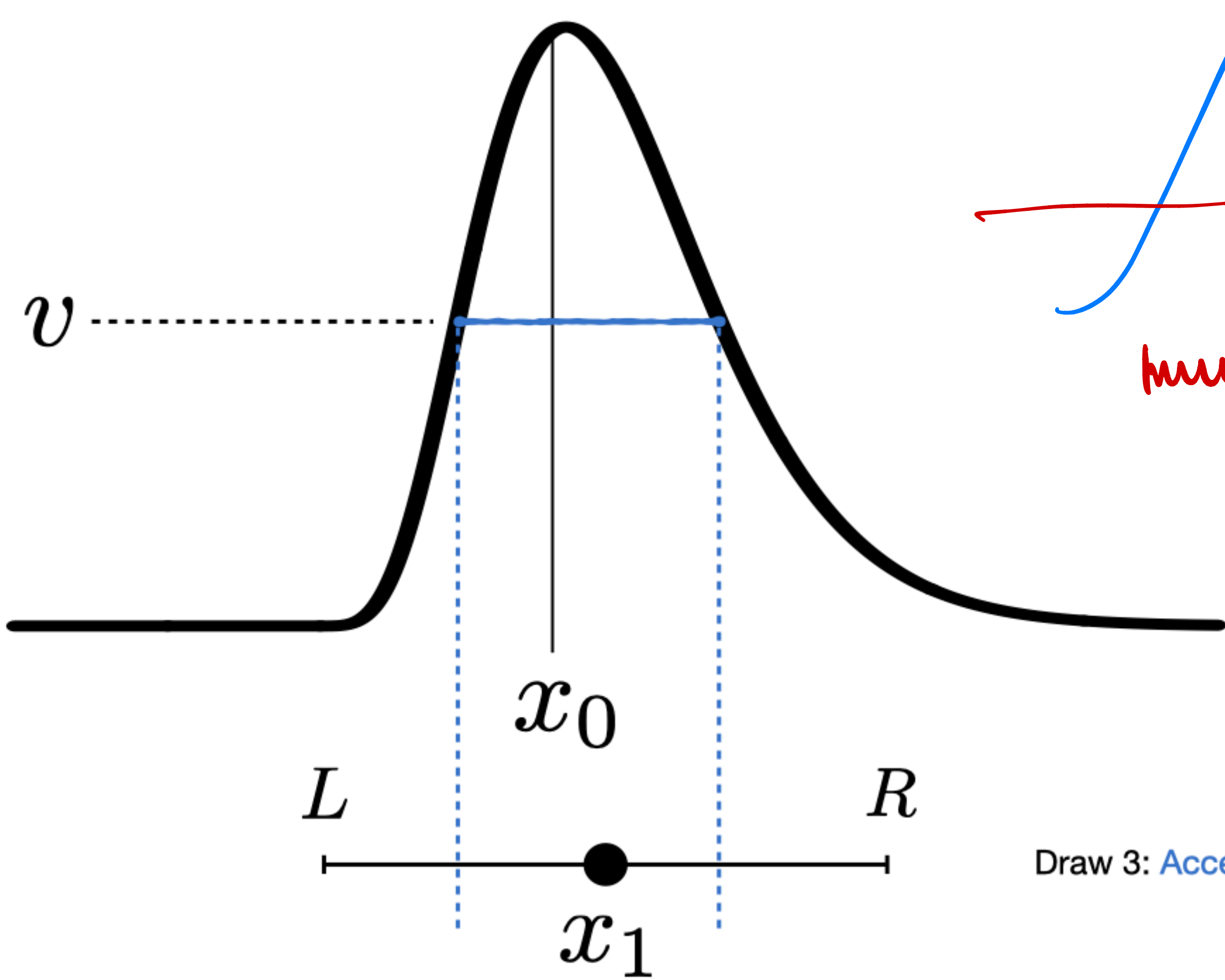
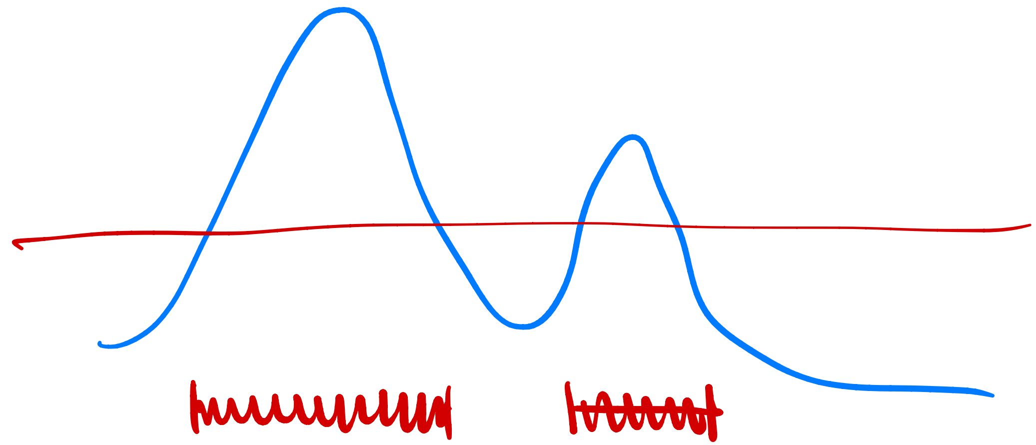


Illustration of shrinkage procedure



Draw 3: Accept



Separated intervals can be reached as long as the modes are not separated by an interval with density 0.

Even this can be overcome if the stepping-out with is large enough.

Use within a Gibbs sampler

Example

Implement a one-at-a-time slice-within-Gibbs sampler for the bioassay example (BDA3, Ch 3.7)

Recall the model: $y_i \overset{\text{ind}}{\sim} \text{Binomial}(n_i,\, \theta_i)$ for $i = 1, \ldots, n$

with $\log\left(\frac{\theta_i}{1-\theta_i}\right) = \alpha + \beta x_i$.

See posted code

Other Methods

Other advanced MCMC methods

- Elliptical slice sampling (Murray et al., 2010; Nishihara et al., 2014)
- Metropolis-adjusted Langevin algorithm (MALA; Wikipedia)
- Hamiltonian Monte Carlo (HMC; BDA3 Ch 12.4-12.5)
- Locally balanced proposals for large discrete sample spaces (Zanella, 2020)
- Others in BDA3 Ch 12

Intractable likelihood

What if we can’t even compute the likelihood?

This occurs with “black box” models that may be simulated, but not explicitly computed.

Examples

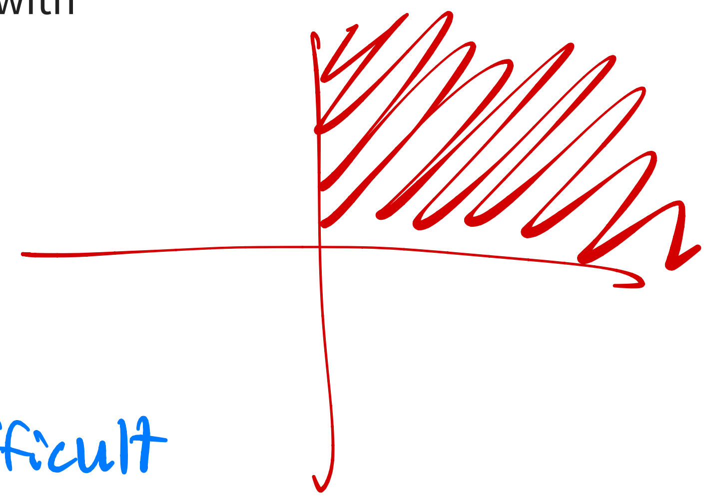
- Complex data-generating models (e.g., climate/earthquake/geology simulation)
- Genetic models
- Models on large discrete spaces (e.g., random partition models)

Intractable likelihood

Example

Consider $(y_1, y_2) \sim \text{Normal}_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})\, 1\{y_1 > 0, y_2 > 0\}$ with

- $\boldsymbol{\mu} \sim \text{Normal}_2(\boldsymbol{m}, \boldsymbol{V})$, and
- $\boldsymbol{\Sigma}$ known.



What is the posterior density for $\boldsymbol{\mu}$?

Extremely difficult to calculate because even the likelihood is difficult to calculate ...

$$f(y_1, y_2 \mid \mu, \Sigma) = \frac{(2\pi)^{-\frac{2}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left[-\frac{1}{2}(y-\mu)^T \Sigma^{-1}(y-\mu)\right] 1\begin{pmatrix} y_1 > 0 \\ y_2 > 0 \end{pmatrix}}{\int_0^\infty \int_0^\infty \text{above} \; dy_1 \, dy_2}$$

Intractable likelihood

Computation methods for posterior sampling:

- Approximate Bayes Computation (ABC; BDA3 Ch 13.9, Wikipedia)
- Pseudo-marginal MCMC (Tutorial and examples, Wikipedia)
- Strategies in BDA3 Ch 13.10